

Let's begin our analysis of this circuit, assuming that the output is high ($= V_{DD}$) and the input is low ($= 0$ V). Figure 18.4 shows the bottom portion of the Schmitt trigger used in calculating the upper switching point voltage, V_{SPH} . MOSFETs M1 and M2 are off, with $V_{in} = 0$ V while M3 is on. The source of M3 floats to $V_{DD} - V_{THN}$, or approximately 4 V for $V_{DD} = 5$ V. We can label this potential V_x , as shown in the figure.

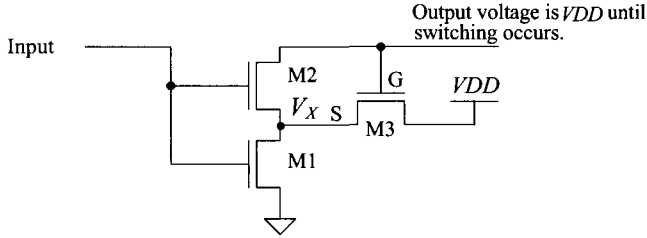


Figure 18.4 Portion of the Schmitt trigger schematic used to calculate upper switching point voltage.

With V_{in} less than the threshold voltage of M1, V_x remains at $V_{DD} - V_{THN3}$. As V_{in} is increased further, M1 begins to turn on and the voltage, V_x , starts to fall toward ground. The high switching point voltage is defined when

$$V_{in} = V_{SPH} = V_{THN2} + V_x \quad (18.2)$$

or when M2 starts to turn on. As M2 starts to turn on, the output starts to move toward ground, causing M3 to start turning off. This in turn causes V_x to fall further, turning M2 on even more. This continues until M3 is totally off and M2 and M1 are on. This positive feedback causes the switching point voltage to be very well defined.

When Eq. (18.2) is valid, the currents flowing in M1 and M3 are essentially the same. Equating these currents gives

$$\frac{\beta_1}{2}(V_{SPH} - V_{THN})^2 = \frac{\beta_3}{2}(V_{DD} - V_x - V_{THN3})^2 \quad (18.3)$$

Since the sources of M2 and M3 are tied together, $V_{THN2} = V_{THN3}$, the increase in the threshold voltages from the body effect is the same for each MOSFET. The combination of Eqs. (18.2) and (18.3) yields

$$\frac{\beta_1}{\beta_3} = \frac{W_1 L_3}{L_1 W_3} = \left[\frac{V_{DD} - V_{SPH}}{V_{SPH} - V_{THN}} \right]^2 \quad (18.4)$$

The threshold voltage of M1, given by V_{THN} in this equation, is the zero body bias threshold voltage ($= 0.8$ V in our long-channel CMOS process and 0.25 V in the short-channel process). Given a specific upper switching point voltage, the ratio of the MOSFET transconductors is determined by solving this equation. A general design rule for selecting the size of M2, that is, β_2 , is to require that

$$\beta_2 \geq \beta_1 \text{ or } \beta_3 \quad (18.5)$$

since M2 is used as a switch.

A similar analysis can be used to determine the lower switching point voltage, V_{SPL} , resulting in the following design equation:

$$\frac{\beta_5}{\beta_6} = \frac{W_5 L_6}{L_5 W_6} = \left[\frac{V_{SPL}}{V_{DD} - V_{SPL} - V_{THP}} \right]^2 \quad (18.6)$$

The following example illustrates the design procedure for a Schmitt trigger.

Example 18.1

Design and simulate a Schmitt trigger using the short-channel CMOS process with $V_{SPL} = 400$ mV and $V_{SPH} = 700$ mV.

We begin by solving Eqs. (18.4) and (18.6) for the transconductance ratios. For the upper switching point voltage,

$$\frac{W_1 L_3}{W_3 L_1} = \left[\frac{1 - 0.7}{0.7 - 0.25} \right]^2 = 0.444 \rightarrow L_1 = L_3 = 1 \text{ and } W_1 = 10, W_3 = 22.5$$

and for the lower switching point voltage,

$$\frac{W_5 L_6}{W_6 L_5} = \left[\frac{0.4}{1 - 0.4 - 0.25} \right]^2 = 1.3 \rightarrow L_5 = L_6 = 1 \text{ and } W_6 = 20, W_5 = 26$$

M2 is set to 10/1 and M4 is set to 20/1. Simulation results are seen in Fig. 18.5. This figure reveals the benefit of using a Schmitt trigger, namely, it allows slow moving inputs to be made into good solid logic high and low values. In a practical circuit, connecting the ramp-shaped input seen in Fig. 18.5 to an inverter would produce oscillations in the inverter's output (because of noise on the ramp). ■

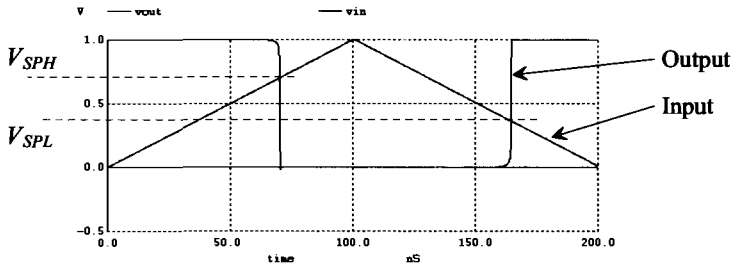


Figure 18.5 The input and output of the Schmitt trigger designed in Ex. 18.1.

Switching Characteristics

The propagation delays of the Schmitt trigger can be calculated in much the same way as the inverter of Ch. 11. Defining equivalent digital resistances for M1, M2, M4, and M5 as R_{n1} , R_{n2} , R_{p4} , and R_{p5} , respectively, gives a high-to-low propagation delay-time, neglecting the Schmitt trigger output capacitance, of

$$t_{PHL} = 0.7 \cdot (R_{n1} + R_{n2}) \cdot C_{load} \quad (18.7)$$

and

$$t_{PLH} = 0.7 \cdot (R_{p4} + R_{p5}) \cdot C_{load} \quad (18.8)$$

18.1.2 Applications of the Schmitt Trigger

Consider the waveform shown in Fig. 18.6. A pulse with ringing is a common voltage waveform encountered in buses or lines interconnecting systems. If this voltage is applied directly to a logic gate or inverter input with a V_{SP} of 0.5 V, the output of the gate will vary with the period of the ringing on top of the pulse. Using a Schmitt trigger with properly designed switching points can eliminate this problem.

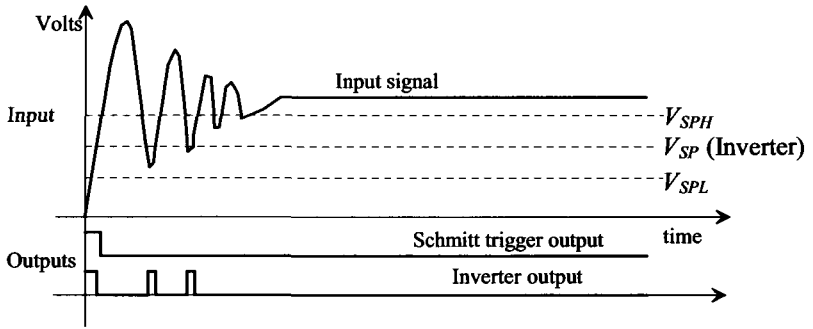


Figure 18.6 Applying a Schmitt trigger to clean up an interconnecting signal.

The Schmitt trigger can also be used as an oscillator (Fig. 18.7). The delay-time in charging and discharging the capacitor sets the oscillation frequency. At the moment in time when the output of the Schmitt trigger switches low, the voltage across the capacitor is V_{SPH} . The capacitor will start to discharge toward ground. The voltage across the capacitor is given by

$$V_c(t) = V_{SPH} \cdot e^{-t/RC} \quad (18.9)$$

At the time when $V_c(t) = V_{SPL}$, the output of the Schmitt trigger changes state. This time is given by solving Eq. (18.9) by

$$t_1 = RC \cdot \ln \frac{V_{SPH}}{V_{SPL}} \quad (18.10)$$

A similar analysis for the case when the capacitor is charged from V_{SPL} to V_{SPH} gives

$$V_c(t) = V_{SPL} + (V_{DD} - V_{SPL}) \left(1 - e^{-\frac{t}{RC}}\right) \quad (18.11)$$

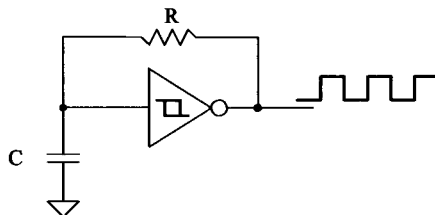


Figure 18.7 Oscillator design using a Schmitt trigger.

and

$$t_2 = RC \cdot \ln \frac{V_{DD} - V_{SPL}}{V_{DD} - V_{SPH}} \quad (18.12)$$

The oscillation frequency, neglecting the intrinsic delay of the Schmitt trigger, is given by

$$f_{osc} = \frac{1}{t_1 + t_2} \quad (18.13)$$

The capacitance used in these equations is the sum of the input capacitance of the Schmitt trigger and any external capacitance.

An alternative oscillator using the Schmitt trigger is shown in Fig. 18.8. Here the MOSFETs M1 and M4 behave as current sources (see Ch. 20) mirroring the current in M5 and M6. When the output of the oscillator is low, M3 is on and M2 is off. This allows the constant current from M4 to charge C . When the voltage across C reaches V_{SPH} , the output of the Schmitt trigger swings low. This causes the output of the oscillator to go high and allows the constant current from M1 to discharge C . When C is discharged down to V_{SPL} , the Schmitt trigger changes states. This series of events continues, generating the square wave output.

If we label the drain currents of M1 and M4 as I_{D1} and I_{D4} , we can estimate the time it takes the capacitor to charge from V_{SPL} to V_{SPH} as

$$t_1 = C \cdot \frac{V_{SPH} - V_{SPL}}{I_{D4}} \quad (18.14)$$

and the time it takes to charge from V_{SPH} to V_{SPL} is

$$t_2 = C \cdot \frac{V_{SPH} - V_{SPL}}{I_{D1}} \quad (18.15)$$

The period of the oscillation frequency is, as before, the sum of t_1 and t_2 .

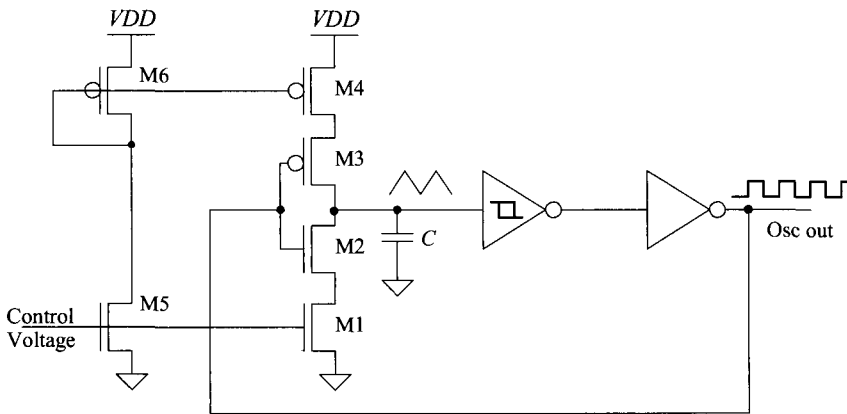


Figure 18.8 Voltage-controlled oscillator using Schmitt trigger and current sources. MOSFETs M2 and M3 are used as switches.

This type of oscillator is termed a voltage-controlled oscillator (VCO) since the output frequency can be controlled by an external voltage. The currents I_{D1} and I_{D4} , (Fig. 18.8) are directly controlled by the control voltage. As we will see in Ch. 20, the current in M5 is mirrored in M1, M4, and M6, with an appropriate scaling factor dependent on the size of the transistors.

18.2 Multivibrator Circuits

Multivibrator circuits (Fig. 18.9) are circuits that employ positive feedback. The name “multivibrator” is a vestige from early-time electronics development (prior to the ubiquitous term “digital”) where the circuits’ outputs vibrate between two states. There are three types of multivibrators: astable, bistable, and monostable. Astable multivibrator circuits are unstable in either output (high or low) state. The oscillators that we have discussed are examples of astable multivibrators. The bistable multivibrator is stable in either the high or low state. Flip-flops and latches are examples of the bistable multivibrator. Monostable multivibrators are stable in a single state. Monostable multivibrators are also called one-shots. In this section we discuss the monostable and astable multivibrators. We distinguish the material in this section from the other material in the book by using resistor-capacitor time constants to set time intervals.

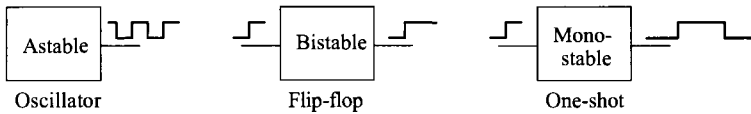


Figure 18.9 Multivibrator circuits.

18.2.1 The Monostable Multivibrator

A CMOS implementation of the monostable multivibrator is shown in Fig. 18.10. Under normal conditions, V_{in} is low and the output of the NOR gate, V_1 , is high. The voltage V_2 is pulled high through the resistor, and the output of the inverter, V_3 , is a low. Upon application of a trigger pulse, that is, V_{in} going high, both V_1 and V_2 drop to zero volts and the output of the inverter, which is also the output of the monostable, goes high. This output is fed back to the input of the NOR gate holding V_1 at ground potential.

After triggering, the potential V_2 will start to increase because C is charged through R . The potential across the capacitor after triggering takes place is given by

$$V_c(t) = V_2(t) - \overbrace{V_1(t)}^{=0} = VDD \cdot (1 - e^{-\frac{t}{RC}}) \quad (18.16)$$

If we assume that the V_{SP} of the inverter is $VDD/2$, then the time it takes for the capacitor to charge to V_{SP} is given by

$$t = RC \cdot \ln \frac{VDD}{VDD - V_{SP}} = RC \cdot \ln(2) \approx 0.7RC \quad (18.17)$$

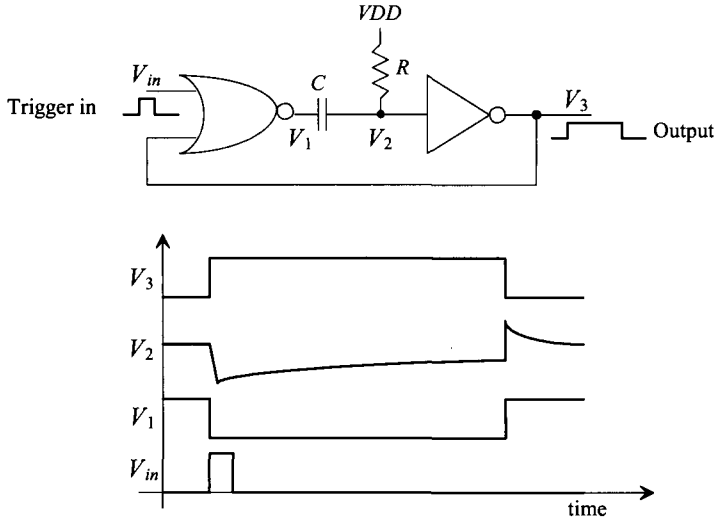


Figure 18.10 Operation of the monostable multivibrator.

This time also defines the output pulse width, neglecting gate delays, since the inverter switches low. The inverter output, V_3 , going low causes V_1 to go back to V_{DD} and V_2 to go to $V_{DD} + V_{DD}/2$. If the resistor or capacitor is bonded out (connected to the output pads), the ESD diodes may keep V_2 from going much above $V_{DD} + 0.7$. The time it takes V_2 to decay back down to V_{DD} limits the rate at which the one-shot can be retriggered. Also note that the trigger input can be longer than the output pulse width. Longer output pulse widths may cause V_2 to go as high as 10 V as well as limit the maximum trigger rate.

18.2.2 The Astable Multivibrator

An example of an astable multivibrator is shown in Fig. 18.11. This circuit has no stable state and thus oscillates. To analyze the behavior of this multivibrator, let's begin by assuming that the output, V_3 , has just switched high. The output going high causes V_1 to go high (to $V_{DD} + V_{SP1}$), forcing V_2 low. The voltage across the capacitor after this switching takes place is given by

$$V_c(t) = V_1(t) - \overbrace{V_3(t)}^{= V_{DD}} = (V_{DD} + V_{SP1}) \cdot e^{-\frac{t}{RC}} - V_{DD} \quad (18.18)$$

The output of the astable will go low, $V_3 = 0$, when $V_1 = V_{SP1}$. Substituting this condition into the previous equation gives the time the output is high (or low) and is given by

$$t_1 = RC \cdot \ln \frac{V_{DD} + V_{SP1}}{V_{SP1}} = t_2 \quad (18.19)$$

If $V_{SP1} = V_{DD}/2$, then

$$t_1 = t_2 = 1.1RC \quad (18.20)$$